Free constructions

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Inspired by Section 1.7, "Free categories", Awodey.

Vague description of the problem:

> Given an object X with algebraic structure A_1, "freely" construct an object Y of algebraic structure A_2.

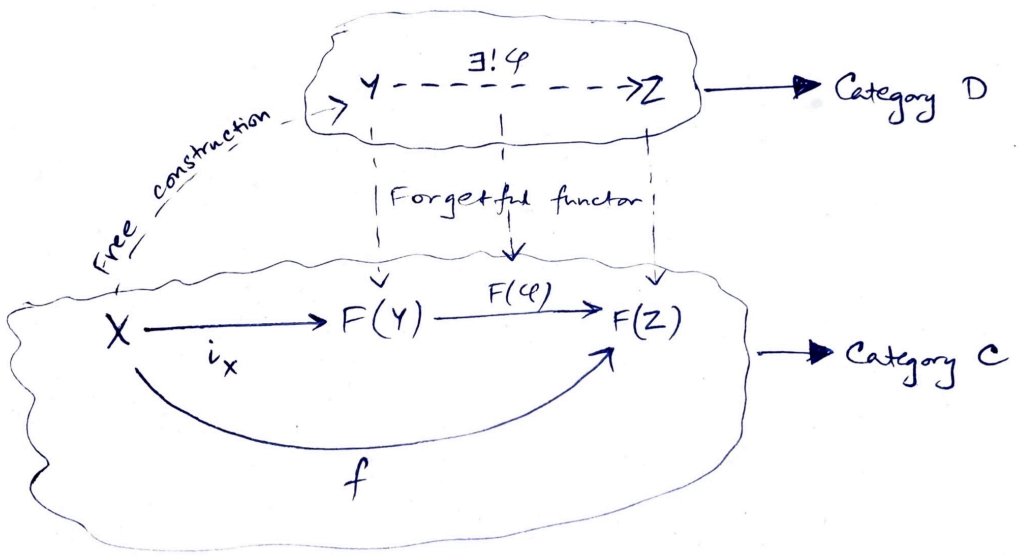
Some instances might be:
- Given a set S, freely construct a group from it.
- Given a set S, freely construct a monoid from it.
- Given a (directed) graph G, freely construct a category out of it.

Attempt to formalize:

> Given categories $\boldsymbol{C}, \boldsymbol{D}$, an object $X \in \mathrm{obj}(\boldsymbol{C})$, and a functor $F : \boldsymbol{D} \to \boldsymbol{C}$, $Y \in \mathrm{obj}(\boldsymbol{D})$ produced by ??? from X is called the "free construction of an object of algebraic structure $\boldsymbol{D}$ from X".
>
> *F is called the forgetful functor from $\boldsymbol{D}$ to $\boldsymbol{C}$.*

Attempt 2:

> Given categories $\boldsymbol{C}, \boldsymbol{D}$, an object $X \in \mathrm{obj}(\boldsymbol{C})$, and a functor $F : \boldsymbol{D} \to \boldsymbol{C}$, we call $Y \in \mathrm{obj}(\boldsymbol{D})$ the "free construction of an object of algebraic structure $\boldsymbol{D}$ from X" iff
>
> $$\exists i_X \in \mathrm{Hom}_C(X, F(Y)), \text{ such that,}$$
> $$\forall Z \in \mathrm{obj}(\boldsymbol{D}), \forall f \in \mathrm{Hom}_C(X, F(Z)), \exists! \varphi \in \mathrm{Hom}_D(Y, Z), f = F(\varphi) \circ i_X$$
>
> Observe that we need i_X alongside Y to define the free construction. To be less symbolically acute, given any $Z \in \mathrm{obj}(\boldsymbol{D})$, and any morphism f (in $\boldsymbol{C}$) from X to $F(Z)$, there exists a unique morphism φ (in $\boldsymbol{D}$) such that $f = F(\varphi) \circ i_X$.



Observe that this means that $\mathrm{Hom}(Y, Z) \cong \mathrm{Hom}(X, F(Z))$, i.e.,

$$\mathrm{Hom}(\mathrm{Free}(X), Z) \cong \mathrm{Hom}(X, \mathrm{Forget}(Z)).$$

Can this be simplified to say that the tuple $(i_X, F(Y))$ is an initial object of the coslice category $X/F(\boldsymbol{D})$? Is there a nice way to define $F(\boldsymbol{D})$ as a subcategory of $\boldsymbol{C}$ to make this work?

If the forgetful functor F was faithful, then this does make sense, because a unique $F(\varphi)$ (produced when the initial object in the coslice category is presented with $(f, F(Z))$) would uniquely identify φ. [Nah, still wrong, the produced function may not be of the form $F(\varphi)$... Looks like a deadend here.]

Or maybe I am considering the wrong category to find initial objects in.

Instead of taking the coslice category, consider the category where objects are tuples $(f, Z)[Z \in \mathrm{obj}(\boldsymbol{D}), f \in \mathrm{Hom}_C(X, F(Z))]$, and morphisms between (f_1, Z_1) and (f_2, Z_2) correspond to morphisms $\psi : Z_1 \to Z_2$ such that $F(\psi) \circ f_1 = f_2$. Initial objects in this category correspond to (i_X, Y).

The Wikipedia article on comma categories mentions that while in the most general case it is defined on two functors S and T which share the same codomain, most of the time one of the two functors has domain $\mathbf{1}$ (the one-object category).

Indeed, the category that we have described above is the comma category $!_X \downarrow F$, where $!_X : \mathbf{1} \to \boldsymbol{C}$, maps the single object in $\mathbf{1}$ to X in $\boldsymbol{C}$. So, a free construction based on the object X corresponds to initial objects in $!_X \downarrow F$. *Have a look at the **Comma categories** page down below.*

An even more natural perspective is through **universal morphisms**. The diagram drawn above is exactly what it means for (Y, i_X) to be a universal morphism from X to the forgetful functor F. Therefore, universal morphisms from X to the forgetful functor correspond to free constructions on X. *Have a look at the **Universal morphisms** page below.*

If we think of X as a set, and Z as a group, then (Y, i_X) being a free group simply means that any "function from X to Z" "uniquely factors through" the function i_X via a homomorphism from Y to Z.

The above wording "identifies" each group with its underlying set. Functions from $X \to Z$ are actually morphisms in $\mathbf{Set}$ from $X \to F(Z)$, and so on.

The above "factoring through" notion fits nicely with the idea of (Y, i_X) being initial in $!_X \downarrow F$ which is described above.

Wikipedia says "Forgetful functors tend to have left adjoints, which are 'free' constructions"(https://en.wikipedia.org/wiki/Forgetful_functor). So this might be a route to learning adjoints.

Read the part "Left adjoints of forgetful functors" on the Wiki page linked above. There is some nice intuition there, also something about a vector space being a free construction on its basis set.